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RESEARCH ARTICLE

Hierarchical Cascade Deep Learning for EMG-Based Behavioral Biometrics: Gesture and Subject Classification

BERKE CANSIZ¹, **HATICE VILDAN DUDUKCU¹**, **MURAT TASKIRAN¹**,
AND NIHAN KAHRAMAN¹

Department of Electronics and Communication Engineering, Yıldız Technical University, 34349 Istanbul, Türkiye

Corresponding author: Berke Cansiz (berke.cansiz@std.yildiz.edu.tr)

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ABSTRACT The integration of electromyography (EMG) signals into biometric recognition has garnered significant attention due to their potential for highly secure and reliable identification. Unlike vision-based methods like cameras, EMG is immune to lighting conditions, clothing, or occlusions. This study presents a hierarchical cascade deep learning framework aimed at simultaneously performing hand gesture recognition and subject-specific biometric classification. Utilizing the publicly available Gesture Recognition and Biometrics ElectroMyogram (GRABMyo) dataset, which encompasses diverse EMG recordings from 43 individuals performing 17 unique gestures, this study proposes a two-staged classification approach. The first stage concentrates on recognizing the hand gesture, succeeded by a gesture-specific model that subsequently categorizes the subject associated with the identified gesture. The experimental results demonstrate the effectiveness of the proposed model, which achieved an average accuracy of 71.62% across gesture and subject classification, representing an improvement of approximately 5% and 21% compared to conventional single-model and multi-task strategies evaluated in this study, highlighting this approach’s effectiveness in handling the variability of EMG signals across different gestures and subjects. The findings underscore the potential of the proposed methodology for enhancing EMG-based biometric recognition systems.

INDEX TERMS Behavioral biometrics, cascade deep learning, electromyography, gesture classification, subject classification.

I. INTRODUCTION

In recent years, the analysis of biomedical signals has emerged as a significant research focus in fields such as human-machine interface, biometric recognition, and prosthetic control systems. Among these signals, Electromyography (EMG) signals, which measure the electrical traces of muscle activity, can play a significant role, particularly in the recognition of hand movements and the authentication of

individuals. Electromyography (EMG) signals are acquired by detecting the electrical outputs of action potentials produced during muscle contractions, which are recorded from the skin surface using electrodes. These signals convey information pertinent to muscle activity, rendering them highly important for movement recognition and identifying unique biometric patterns. In the context of biometrics, EMG presents a promising alternative to traditional modalities such as fingerprints or facial recognition, as it is intrinsically linked to internal neuromuscular activity, making it more challenging to replicate or spoof. Additionally, the variability

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of EMG signals across individuals offers a rich source of distinctive features that can be used for secure and reliable personal identification.

This study integrates two main objectives: hand gesture recognition and person-specific biometric recognition, by utilizing a hierarchical cascade deep learning methodology. The proposed method first identifies the hand gesture for the input signal, followed by determining the subject to whom the signal belongs using a gesture-specific classification model.

In this study, the publicly available Gesture Recognition and Biometrics ElectroMyogram (GRABMyo) Dataset was utilized to perform both gesture and subject classification using surface electromyography (sEMG) signals [1], [2], [3]. The dataset comprises EMG recordings from 43 people, each executing 17 unique hand motions, gathered during three independent sessions, resulting in a total of 129 session recordings. Each gesture sample was concatenated from all EMG channels and fed into a CNN model designed for multi-class classification across 17 gesture types. A gesture-based subject classification was then applied, using a different subject classifier model for each gesture resulting in a hierarchical cascade structure.

This study introduces a hierarchical cascade design that sequentially executes gesture recognition and gesture-specific subject classification, a methodology rarely used in EMG-based biometric systems. Moreover, employing distinct subject classification models for each gesture addresses the variability in EMG signals in different motions, which frequently impairs overall classification efficacy.

The remainder of this paper is organized as follows: the next subsection reviews the related work in EMG-based gesture and biometric recognition and explains the contribution of this study. Section II details the proposed model architecture and evaluation metrics. Section III describes the dataset, simulation setup, and analysis of the classification results, while Section IV discusses the implications of the findings and outlines future directions for this research.

A. RELATED WORKS

The analysis of electromyography (EMG) data for subject identification and hand gesture recognition has garnered a lot of interest in the fields of biomedical signal processing and human-computer interaction in recent years. EMG-based systems provide a affordable, and non-invasive way to improve rehabilitation procedures, provide biometric authentication, and control prostheses. Numerous studies have explored the use of machine learning and deep learning techniques for gesture classification using multi-channel EMG data.

Conventional methods mostly employed feature extraction in conjunction with traditional classifiers like random forests [4] or support vector machines (SVM) [5]. More recently, Convolutional Neural Networks (CNNs) [6], [7], [8] and recurrent neural networks (RNNs) [9] such as gated recurrent unit (GRU) [10] and long short-term memory

(LSTM) [11] have demonstrated superior performance by enabling end-to-end learning directly from raw or preprocessed EMG signals.

Recent advances in EMG-based gesture recognition have significantly expanded the potential of human-computer and human-robot interactions, particularly in assistive technologies. One direction of progress involves multimodal fusion methods, as demonstrated by [12], where transient-state signal sEMG and Force Myography (FMG) signals were fused using a cross-attention convolutional neural network (CNN-ATT) model. This method surpassed single-modality systems, with a gesture classification accuracy of 96%, so confirming the effectiveness of combining modalities for improved recognition accuracy. In a related work, [13] proposed integrating EMG and voice recognition within a machine learning framework for controlling a prosthetic hand with six degrees of freedom. Their method attained 94.8% accuracy in voice commands and 76% in EMG classification, allowing intuitive control of five distinct hand gestures. In addition to these methods, [14] proposed an approach that transforms sEMG signals in the time domain into grayscale images, enabling CNN-based classification without manual feature extraction. The model trained with these images reached 98.03% accuracy on custom data and 99.95% and 98.07% accuracy for two benchmark datasets respectively.

Alongside advancements in modality and data representation, several research has concentrated on enhancing deep learning model architectures to improve classification performance. For example, [15] proposed the multivariate variational mode decomposition and the two-channel convolutional neural network with added attention mechanism (MVMD-TCCA), which fuses multivariate variational mode decomposition with attention-based CNNs. This method achieved 97.90% accuracy on their custom dataset and demonstrated a significant improvement over the classification of raw signals. As an alternative method, [16] presented the SEMG-based gesture recognition network (SGRN), a multi-branch spatial-temporal network consisting of spatial, temporal, and spatial-temporal networks that simultaneously extract and integrate information, surpassing previous state-of-the-art models for four datasets. In addition, these studies highlight the variety of approaches used to improve EMG-based gesture detection, including multimodal signal fusion, visual signal encoding, and attention-enhanced deep networks. The findings indicate a strong trend towards enhanced accuracy, and user-friendliness in systems designed for practical application in assistive technologies.

EMG signals uniquely integrate physiological and behavioral characteristics, making them appropriate for high-security biometric applications [17]. Especially with two-stage authentication structures, it is enable the development of robust verification systems that effectively combine password and EMG characteristics. In one of these structures, [18] initially verified the passcode through a specific sequence of finger movements (for instance, contractions

of the index, thumb, and middle fingers in a designated order) utilizing EMG signals captured from the hand's surface; subsequently, the verification process advanced by analyzing the EMG characteristics of users who successfully completed this stage. This approach effectively ensured that an unauthorized user could not gain access to the system, even if they had acquired the password. The system they established reported an Equal Error Rate (EER) value of 0.0128. In another study presented in [19], a disentanglement model was introduced, which separates the global EMG representations into password-specific and biometric-specific components in two distinct latent spaces using a multibranch encoder-decoder architecture. This approach achieved an Equal Error Rate (EER) of 0.0075 when impostors did not possess knowledge of the password. Similarly [20] presented a feature disentanglement technique that separates sEMG features into pattern-specific and subject-specific components, resulting in improved accuracy for cross-subject and cross-session gesture classification. In another study [21], person recognition with EMG data was performed using multi-code structure with decision level fusion. Accordingly, candidate persons were determined for each gesture for a certain number of movements; then, these candidates were combined at the decision level to obtain the final person identification. This method achieved a notably low False Reject Rate (FRR) of approximately 0.01%. Meanwhile, [22] demonstrated that even a single EMG channel could effectively identify users for a smaller group of people, achieving an F1-score of 96.40% using a stacked Long Short-Term Memory (LSTM) neural network and making the system more practical for real-world implementation. Additionally, to address gesture-independence, [23] proposed a deep learning model gradient-weighted class activation mapping (Grad-CAM), that identifies users by decoding and visualizing the unique user-specific neurophysiological patterns. This method achieved 97% accuracy when trained and tested on the same gestures and 90% when gestures vary from training. In a complementary direction, [24] used bag-of-words (BoWs) classification models using FFT-based features in six nonoverlapping frequency bands. This method achieved nearly perfect AUC and EER in within-day scenarios and showed minimal degradation across days, maintaining over 99% Rank-5 accuracy in cumulative match characteristics. These recent studies show that EMG-based biometric systems are becoming a growing field of research. However, since the time-sensitive nature of biometric data, it is crucial to systematically analyze the performance of EMG signals over time [17]. In fact, a study [25] reported that the accuracy 93% was achieved in the Rank-5 classification to determine the right person among five candidate users in evaluations conducted over multiple days. In addition to achieving high accuracy in biometric system designs, low memory requirements and fast inference time are also critical. In this direction, in a study conducted by Zhang et al. [26], inference could be achieved in an average time of 227.76 ms,

while the system also exhibited high performance with an accuracy rate of 98.7%. Similarly, in another study [27], a lightweight CNN model with only 34,311 parameters was used to achieve 98.81% accuracy. Collectively, these studies highlight that EMG/sEMG-based biometric systems offer a high degree of accuracy, robustness, and adaptability, demonstrating strong potential for reliable deployment in real-world authentication and identification scenarios. When these systems are evaluated, it is seen that features such as multi-day durability, lightweight model structure, fast extraction time and high accuracy constitute the ideal performance criteria for EMG-based biometric systems.

Despite these advancements, gesture recognition and subject classification have predominantly been treated as separate tasks in prior research. While the use of EMG for biometric identification is gaining attention, it remains a relatively underexplored area. Moreover, existing studies often rely on subject-level models trained across all gesture types, which can obscure the unique signal characteristics associated with individual gestures, thereby limiting the potential for capturing gesture-specific biometric features.

B. CONTRIBUTION AND OUTLINE

In order to advance prior research, the key contributions of this study are outlined as follows.

- Most of the studies applied with the existing dataset have focused, either solely on gestures or subject classification only. In the proposed study, the two types of classification were combined with the application of the hierarchical cascade, and an attempt was made to classify both the subject and the gesture applied by the subjects.
- In the current literature, biometric subject classification studies are carried out including all gestures applied by person, and this situation may affect classification performance due to the dominance of signal pattern differences between gestures. For this reason, in the proposed study, a separate person classification was performed for each gesture. In this way, the detection of high and low discrimination movements between people was ensured.
- The architecture of the presented hierarchical cascade model was created with low parameter load using point-wise and depth-wise convolution structures, which contributes to low inference latency.

II. METHODOLOGY

In the proposed study, two distinct types of model structures were implemented: one for gesture classification and another for subject classification. A cascade structure was developed to integrate these two tasks as described in Figure 1. In this approach, an incoming signal is first processed so that the gesture can be identified. Once the gesture has been determined, the subject classification model associated with

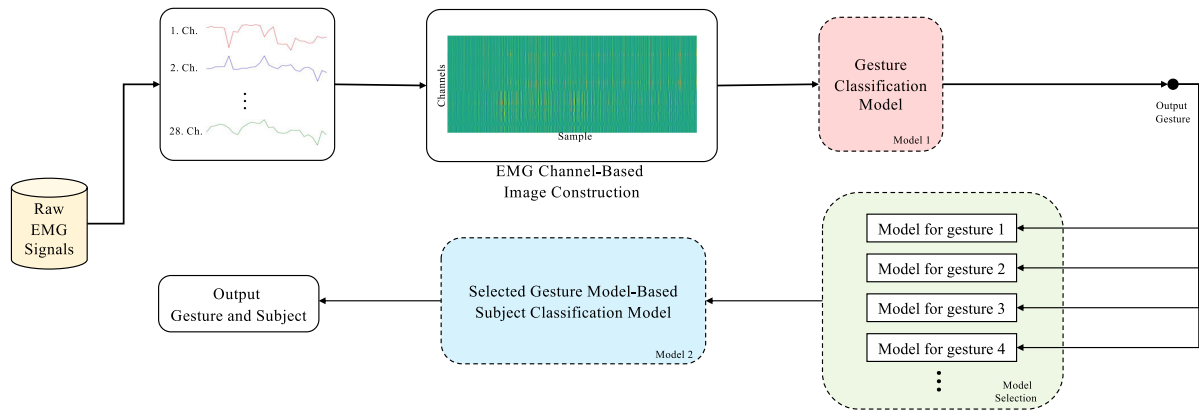


FIGURE 1. Block diagram of proposed methodology with cascaded deep learning models.

that gesture is selected. Finally, the specific subject present in the signal is determined using the selected model.

A. CONVOLUTIONAL NEURAL NETWORK

CNN is a deep learning model with an architecture that automatically performs feature extraction and classification on image and signal data. One of its earliest applications was the classification of handwritten digits in the late 1990s using the LeNet model [28]. Later, AlexNet became popular by achieving success with its multilayer architecture in the ImageNet dataset in 2012 [29]. The architecture of a basic CNN model consists of successively structured convolutional layers, pooling layers, batch normalization applications, and various nonlinear activation functions.

In convolutional layers, various filtering operations are performed on the input data. This filtering operation is carried out using sliding windows with various matrices, called kernels, which have learnable parameters. At the output of this layer, the input data is transformed into filtered data, repeated according to the number of selected filters. The resulting stack of filtered data produced by the convolutional layer is termed the feature maps of the inputs. While feature maps in the initial layers represent basic features such as edges and boundaries of objects, they capture more complex features of the input as the depth of the network increases. Various types of convolutional layers have recently been employed more efficiently in recent architectures. One of these layer structures is the Depthwise convolutional layer [30]. In this approach, instead of filters that process between channels in the basic convolutional layers, a separate and learnable filter is used for each channel; thus, each channel is processed independently. This approach effectively reduces the model size—specifically, the number of parameters, by replacing the multidimensional kernel structure that operates across channels. During a depthwise convolution setup, groups should be formed according to the number of channels in order to ensure that each input channel can be processed independently; so each group only processes using the filter for its own channel. Another

convolutional layer structure is the pointwise approach [30]. In this layer structure, point filters are used between channels to ensure information transfer between channels. Using these pointwise kernels (1×1), instead of spatial information, information is transferred between channels, and maps are synthesized based on this transfer. In general use, this layer is implemented after the depthwise layer that extracts spatial information, in order to ensure information transfer in feature maps belonging to separate channels. With this cooperation of these two approaches, it becomes possible to perform spatial and channel analysis of the input in an optimized way.

In subsequent steps of the basic CNN model, different functions are also used to feed the layers in an optimized way. One of these is the pooling layer, which is generally used to reduce computational load. In addition, it mitigates the effects of the curse of dimensionality problem [31], by summarizing the data with a smaller representation. Another used function is batch normalization, which allows the model layer to adapt to new batches more easily, which also allows more stable model [32]. Finally, activation functions enhance the training of the model by capturing non-linear relationships between layers. The most common activation function is the Rectifier Linear Unit (ReLU), which provides ease of operation during backpropagation, by the negative values being returned to 0 and the remaining values being left as they are. Although this function provides ease of operation, it leads to the dying ReLU problem [33] after a while in training models with a lot of negative output. To solve this problem, Exponential Linear Unit (ELU) [34], which makes Exponential reduction on the negative side, and Scaled ELU (SELU) [35], which is the scaled version of this method, are used. Based on preliminary experiments, SELU provided better training stability and convergence compared to ELU and ReLU, especially in our lightweight architecture, making it a more suitable choice for the architecture. Thus, both ease of operation is provided and more stable training is provided. The dropout function, which allows for the closure of random neurons at a certain level and allows the discovery of different connections to prevent the model from memorizing training data and producing

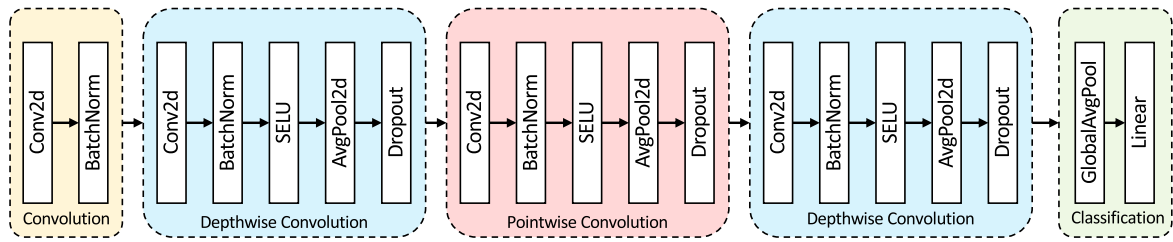


FIGURE 2. Implemented CNN model architecture.

unsuccessful results in test data, is also available in CNN models. A feature vector emerges in the last layer of CNN and this vector is classified using a multi-layer perceptron network.

B. PROPOSED MODEL ARCHITECTURE

CNN models consist of convolution layers; pooling, batch normalization, non-linear activation, dropout functions, and classifier layers. These layers can be combined in different variety to create distinct model architectures. Therefore, in this manuscript, these layers are set up in a certain order and the architecture is designed. Accordingly, the first block contains a basic 2D Convolutional layer and a batch normalization layer. The number of channels increases with this layer. The following block contains a 2D depthwise convolution using $1 \times ch$ kernels, where ch represents the number of input EMG channels (this setting was modified according to EMG configurations, and the corresponding channel numbers are detailed in subsequent sections). followed by batch normalization SELU activation, average pooling and finally dropout function. The relationship between the channels of the signals analyzed with this block is resolved. In the next step, a structure similar to that of the second block was used, this time with 16×1 kernels for processing in the x-plane. This preference aims to extract localized features along the temporal dimension, which is effective for detecting short-term variations in the signal. In the last step, the global average pooling function and a classification layer were added. This architecture aimed to a lightweight design and facilitated both spatial and temporal signal analysis. The architecture of the building model was demonstrated in Figure 2.

C. EVALUATION METRICS

In the literature, a variety of evaluation metrics have been employed to assess the performance of the models developed. Among these, one of the most commonly adopted metrics is the accuracy metric used in general classification success measurements and is calculated by the ratio of correct predictions to all predictions. However, the accuracy metric may provide a misleading indication of the model's actual performance in imbalanced datasets, especially when one class is dominant. Therefore, Precision and Recall metrics, which were developed specifically for such imbalanced

TABLE 1. Gesture classes in GRABMyo dataset.

Gesture ID	Gesture Name
1	Lateral prehension
2	Thumb adduction
3	Thumb and little finger opposition
4	Thumb and index finger opposition
5	Thumb and little finger extension
6	Thumb and index finger extension
7	Index and middle finger extension
8	Little finger extension
9	Index finger extension
10	Thumb extension
11	Wrist flexion
12	Wrist extension
13	Forearm supination
14	Forearm pronation
15	Hand open
16	Hand close
17	Rest

datasets, are other metrics that are used frequently. Given that these two metric representations occupy considerable space in tables and their comparison can be challenging - particularly in cases involving even-numbered evaluations - consolidating them into a single metric facilitates clearer and more efficient analysis. Therefore, the F1 score, which represents the harmonic mean of these two metrics (precision and recall), was also the preferred evaluation metric. The equations of these metrics provided on the confusion matrix are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

where, TP represents True Positive; TN represents True Negative; FP represents False Positive; FN represents False Negative values in confusion matrix.

During the evaluation of the models, separate experimental stages were performed for each session, and these analyzes were performed with the leave one session out strategy,

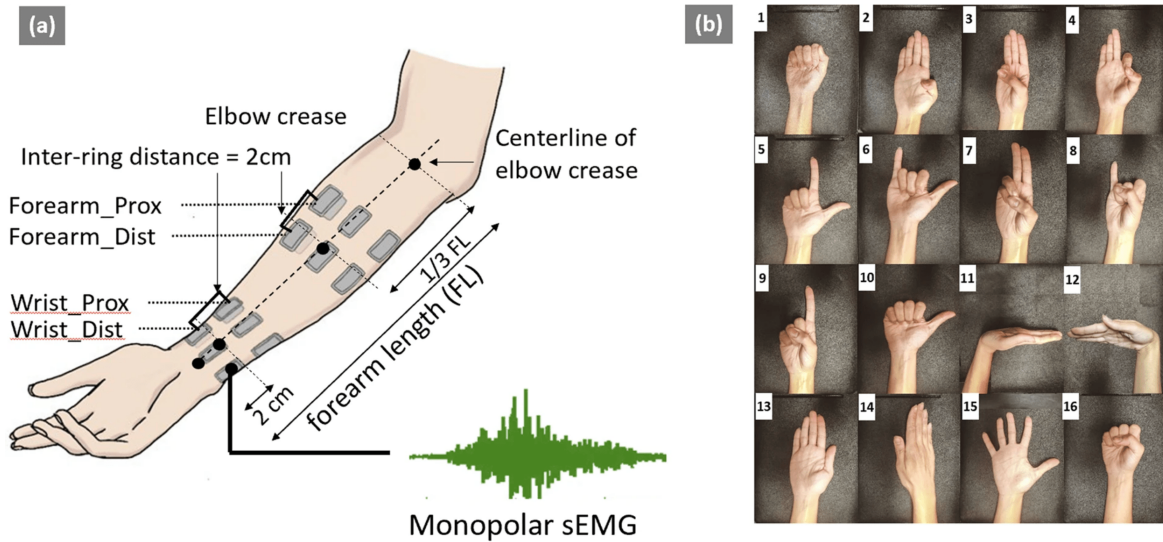


FIGURE 3. (a) Positions of the EMG electrodes. (b) Images of the defined gestures. (The figure is reused from [2], under the Creative Commons Attribution 4.0 International License, without changes.)

and these results were presented with the average, standard deviation, maximum, and minimum values.

III. SIMULATION SETUP AND MODEL DEVELOPMENT

This section provides a detailed description of the dataset employed in the simulation, followed by classification models developed in accordance with the approaches outlined in the methodology section, and concludes with an overview of the overall simulation setup.

A. DATA DESCRIPTION

In the proposed study, the open-access Gesture Recognition and Biometrics ElectroMyogram (GRABMyo) Dataset was used [2], [36]. This dataset includes electromyography (EMG) signals corresponding to 17 different gestures given in Table 1, collected from 43 individuals in 129 recordings in three days (Ethical approval and consent were obtained by the original authors). Throughout these days, the same participants were instructed to repeat their activities at varying intervals, each repetition being recorded as a separate session in the dataset. The signals were obtained from two regions, the wrist and the forearm, with 12 and 16 channels, respectively, totaling 28 channels. The placement of the relevant channels and images of the gestures are demonstrated in Figure 3. The dataset comprises 23 male and 20 female participants, with an average age of 26.35 ± 2.89 years and an average forearm length of 25.15 ± 1.74 cm.

B. PROPOSED MODEL IMPLEMENTATION

The multi-channel structure of EMG signals prevents them from being used directly to train a model in machine learning methods. For this reason, many approaches extract the signals belonging to these channels with the feature engineering method and create the input that will feed their models.

TABLE 2. Gesture classification evaluation results for each session.

	Accuracy	Precision	Recall	F1-Score
Session 1	0.8996	0.9032	0.8996	0.8991
Session 2	0.9035	0.9095	0.9035	0.9041
Session 3	0.8986	0.9054	0.8986	0.8994
Average	0.9006	0.9060	0.9006	0.9009
Std.Dev.	0.0026	0.0032	0.0026	0.0028
Maximum	0.9035	0.9095	0.9035	0.9041
Minimum	0.8986	0.9032	0.8986	0.8991

Among these feature extraction methods, CNN models offer an approach to train the model with image and signal features, especially with the feature extraction part. In this method, various filters extract meaningful information from the data with the convolution process and increase the importance of the features in each training round in order to increase the classification performance, thus providing an overall performance increase. When each channel of multi-channel signals such as EMG signals is aligned along the y-plane and sample points are placed on the x-plane, a sample representation image is obtained. These images are then fed to the two-dimensional CNN model, making it possible to examine the relationships between the channels in detail. The model architecture was designed accordingly including 3 different types of convolution layers - Basic, depthwise and pointwise - and a single classification layer (detailed model design was explained in Section II-B), the architecture of this network model is depicted in Figure 2.

1) GESTURE CLASSIFICATION

The signals obtained from each subject are classified according to the type of gesture to which it is connected. The model architecture adopted in this classification is based on the architecture depicted in Figure 2, and the output

TABLE 3. Accuracy scores of subject classification for each gesture model.

	Gesture 1	Gesture 2	Gesture 3	Gesture 4	
Session 1	0.7076	0.8439	0.7940	0.7907	
Session 2	0.8106	0.8970	0.8306	0.8007	
Session 3	0.7375	0.8538	0.7741	0.8140	
Average	0.7519	0.8649	0.7996	0.8018	
Std. Dev.	0.0530	0.0282	0.0287	0.0117	
Maximum	0.8106	0.8970	0.8306	0.8140	
Minimum	0.7076	0.8439	0.7741	0.7907	
	Gesture 5	Gesture 6	Gesture 7	Gesture 8	
Session 1	0.7708	0.7841	0.8140	0.8040	
Session 2	0.8671	0.8339	0.8904	0.8439	
Session 3	0.8140	0.8671	0.7508	0.8106	
Average	0.8173	0.8284	0.8184	0.8195	
Std. Dev.	0.0482	0.0418	0.0699	0.0214	
Maximum	0.8671	0.8671	0.8904	0.8439	
Minimum	0.7708	0.7841	0.7508	0.804	
	Gesture 9	Gesture 10	Gesture 11	Gesture 12	
Session 1	0.7508	0.8239	0.7508	0.7741	
Session 2	0.8339	0.8239	0.7741	0.8904	
Session 3	0.8538	0.8239	0.7841	0.8372	
Average	0.8128	0.8239	0.7697	0.8339	
Std. Dev.	0.0546	0.0000	0.0171	0.0582	
Maximum	0.8538	0.8239	0.7841	0.8904	
Minimum	0.7508	0.8239	0.7508	0.7741	
	Gesture 13	Gesture 14	Gesture 15	Gesture 16	Gesture 17
Session 1	0.8140	0.8372	0.7309	0.7674	0.4419
Session 2	0.8571	0.8372	0.8173	0.8870	0.5050
Session 3	0.7874	0.8538	0.8040	0.7841	0.4319
Average	0.8195	0.8427	0.7841	0.8128	0.4596
Std. Dev.	0.0352	0.0096	0.0465	0.0648	0.0396
Maximum	0.8571	0.8538	0.8173	0.8870	0.5050
Minimum	0.7874	0.8372	0.7309	0.7674	0.4319

dimension of the final layer was set to 17, corresponding to the number of gestures.

2) SUBJECT CLASSIFICATION

In this section, subject classification was performed to determine which subject the EMG signals belong to. However, there is a potential difficulty in this classification between different gestures. Consequently, different gestures belonging to the same person were predicted to force the model to identify that person and the model would not be able to learn. For this reason, separate training models were created for each gesture in subject classification, and therefore 17 different model trainings were performed.

3) HIERARCHICAL CASCADE DEEP LEARNING FOR GESTURE AND SUBJECT CLASSIFICATION

In the last step, the hierarchical cascade structure was utilized. Although this model does not contain a training process, this model was established by combining the predictions of the models created in the previous step. Accordingly, first, the gesture to which the incoming EMG signal belongs is determined by the first model and according to this output, the subject classification model of the relevant gesture model is selected in the second

model structure completely automatically. As a result of this process, two different outputs (gesture and subject) are obtained. The block diagram of this model is visualized in Figure 1.

C. COMPARISON MODEL IMPLEMENTATION

In the models established to compare strategies, the idea of performing gesture and person classification simultaneously was utilized. In this context, two different model strategies were developed and the details of the model are explained in the following subsections.

1) SINGLE MODEL

Gestures belonging to each subject are in a separate feature space, therefore subjects and gestures belonging to people can be defined as separate classes. In this way, a person and which gesture that person performs can be determined with a single classification model. The construction of such a structure requires creating class definitions in the form of person and gesture. This definition was made in this training strategy. The CNN model used in this training is implemented the same as in the previous architecture, and training was carried out with 43 people and 17 gestures for each in the last layer, with a total of 731 class information.

2) MULTI-TASK MODEL

Gesture and person classification can also be performed by obtaining two different outputs from a single input of a signal. Thus, instead of training two different models, common information transfer up to a certain layer can be created in the last layer, and customized layers can be created for both person and gesture classification. Based on this idea, the multi-task model was created as an alternative baseline classification to the hierarchical cascade model proposed in this study. The architecture used in this method is exactly the same as in Figure 2, and two of the last classification layers were created. In this way, both 43 people and 17 gesture classification processes were performed in parallel.

D. SIMULATION SETUP

In the experimental setup, the ‘leave-one-session-out’ strategy was employed, wherein two sessions were used for training and one session was reserved for testing. Accordingly, before the data separated for sessions, all the EMG signals belonging to the forearm and wrist channels were combined and the relevant gesture was defined with 28-channel data in each example for the gesture classification to be conducted in the first experiment. After this process, the images created with 28 channels and 10240 sample numbers along the x and y axes were organized to feed 2-dimensional convolutional layers. Then, the model architectures constructed in the study were fed with this data. Classification training was performed using the Adam optimizer with a learning rate of 0.001 for 200 epochs. The gesture classification model obtained as a result of the training was saved and stored for later use. Cross-validation was applied separately for each session and thus 3 separate gesture classification models were obtained. In subsequent steps of the experiment, a person classification model was constructed for each gesture. For this, the leave one session out strategy was used again and each gesture data was also separated. Thus, 17 separate sub-data that will perform subject classification for each gesture data were obtained. Then, the training of these 17 models was carried out sequentially for 200 epochs with 0.001 learning rate with Adam optimizer. At the end of 3 session training, a total of 51 models were obtained and the parameter information of these models was also stored. In the last stage, a cascade structure was constructed and a 731-class prediction environment was created. Accordingly, with the first model output, the parameter file of the relevant gesture is selected from among the 17 models and then this parameter information is loaded into the current model. After this process, a subject classification with 43 classes is made and the obtained gesture and subject labels are compared with the ground truth. In this comparison, since correctly identifying both the gesture and the subject is crucial for system security in real-world applications, accuracy is calculated based on the simultaneous correctness of both gesture and subject classification. A prediction contributes to

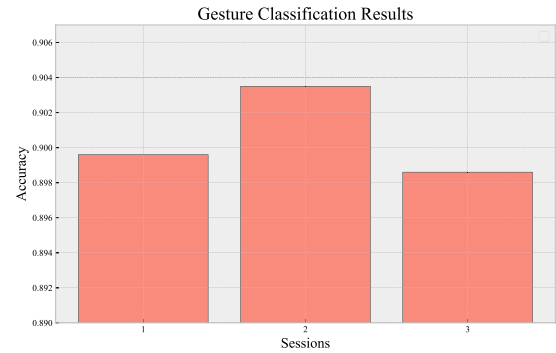


FIGURE 4. Average gesture classification results for each session.

TABLE 4. Results of gesture classification for each session.

	Accuracy	Precision	Recall	F1-Score
Session 1	0.6918	0.6935	0.6918	0.6584
Session 2	0.7457	0.7583	0.7457	0.7213
Session 3	0.711	0.7223	0.711	0.6836
Average	0.7162	0.7247	0.7162	0.6878
Std.Dev.	0.0273	0.0325	0.0273	0.0317
Maximum	0.7457	0.7583	0.7457	0.7213
Minimum	0.6918	0.6935	0.6918	0.6584

the accuracy metric only if both labels are correctly identified; otherwise, it is counted as an incorrect prediction. Finally, the results obtained for all three sessions are presented with evaluation metrics.

IV. RESULTS AND DISCUSSION

In this section, the results of the simulation were presented and discussions were made in line with these results.

A. RESULTS

As a result of the experiment, various result tables were obtained for 3 different classification scenarios, in addition to the results where each session was defined as test data. These results are mentioned in the following subsections.

1) GESTURE CLASSIFICATION RESULTS

According to Table 2, where 17 different gesture analyses were performed on the entire data set, it was observed that 0.9006 (± 0.003) accuracy, 0.9060 (± 0.03) precision, 0.9006 (± 0.003) Recall and 0.9009 (± 0.003) F1-score were obtained when the average results were examined. According to these results (see Figure 4), the highest accuracy performance was 0.9035 in the session-2 data, and the lowest accuracy was 0.8996 in the scenario where session 1 data were used as a test.

2) SUBJECT CLASSIFICATION RESULTS

When the subject classification results of 43 classes created for each gesture were examined from Table 3, it was observed that the average accuracy performance of gesture 2 was 0.8649 (± 0.03). Following this model, gesture 14 was obtained with 0.8427 (± 0.01) accuracy, and gesture 12 was obtained with 0.8339 (± 0.06) accuracy.

TABLE 5. Results of gesture and subject classification for each classification baselines.

		Wrist Channels		Forearm Channels		All Channels	
		Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score
Multi Task Model	Session 1	0.3541	0.3117	0.4464	0.4056	0.5312	0.4880
	Session 2	0.3332	0.2933	0.4774	0.4318	0.5310	0.4853
	Session 3	0.3483	0.3013	0.3709	0.3159	0.4454	0.4036
	Average	0.3452	0.3021	0.4316	0.3844	0.5025	0.4590
	Std.Dev.	0.01	0.01	0.05	0.06	0.05	0.05
Single Model	Session 1	0.4854	0.4457	0.5478	0.5161	0.6447	0.6090
	Session 2	0.5345	0.4959	0.6631	0.6299	0.6863	0.6626
	Session 3	0.4903	0.4556	0.5235	0.4833	0.6746	0.6569
	Average	0.5034	0.4657	0.5781	0.5431	0.6685	0.6428
	Std.Dev.	0.03	0.03	0.07	0.08	0.02	0.03
Cascade Model	Session 1	0.4964	0.4573	0.5880	0.5560	0.6918	0.6584
	Session 2	0.5609	0.5226	0.6619	0.6270	0.7457	0.7213
	Session 3	0.5081	0.4755	0.5900	0.5526	0.7110	0.6836
	Average	0.5218	0.4851	0.6133	0.5785	0.7162	0.6878
	Std.Dev.	0.03	0.03	0.04	0.04	0.03	0.03

TABLE 6. Backbone architecture and number of trainable parameter commonly used in the feature extraction stage of all baseline models.

Layer (Type)	Output Shape	# Trainable Parameters
Conv2d (1)	$[N, 16, 10241, 28]$	1 024
BatchNorm2d (2)	$[N, 16, 10241, 28]$	32
Conv2d (3)	$[N, 64, 10241, 1]$	1 792
BatchNorm2d (4)	$[N, 64, 10241, 1]$	128
SELU (5)	$[N, 64, 10241, 1]$	0
AvgPool2d (6)	$[N, 64, 2560, 1]$	0
Dropout (7)	$[N, 64, 2560, 1]$	0
Conv2d (8)	$[N, 64, 2560, 1]$	4 096
BatchNorm2d (9)	$[N, 64, 2560, 1]$	128
SELU (10)	$[N, 64, 2560, 1]$	0
AvgPool2d (11)	$[N, 64, 640, 1]$	0
Dropout (12)	$[N, 64, 640, 1]$	0
Conv2d (13)	$[N, 128, 625, 1]$	2 048
BatchNorm2d (14)	$[N, 128, 625, 1]$	256
SELU (15)	$[N, 128, 625, 1]$	0
AvgPool2d (16)	$[N, 128, 156, 1]$	0
Dropout (17)	$[N, 128, 156, 1]$	0
AdaptiveAvgPool1d (18)	$[N, 128, 1]$	0
Total		9,504

The highest accuracy value was obtained with gesture 2 (see Figure 5) session 2 and this value was 0.897. The lowest average classification result obtained was gesture 17 with 0.505 (± 0.04) accuracy. Following this, gesture 1 and gesture 11 data were 0.7519 (± 0.05) and 0.7697 (± 0.02), respectively. The lowest accuracy score of gesture 17 was 43.19%.

3) SUBJECT AND GESTURE CLASSIFICATION

In this evaluation (see Table 4) where 43 subject analyses were made with 17 different gestures on the entire data set and a total of 731 classes, it was observed that 0.7162 (± 0.03) accuracy, 0.7247 (± 0.03) precision, 0.7162 (± 0.03) Recall and 0.6878 (± 0.03) F1-score were obtained when the average results were examined. The highest accuracy performance was 0.7457 in the session-2 data, and the lowest accuracy was

TABLE 7. Architectures of the baseline models and their respective number of trainable parameters. (The backbone structure used across all models corresponds to the configuration detailed in Table 6.)

Multi Task Model		
Layer (Type)	Output Shape	# Trainable Parameters
Feature Extraction Backbone (1–18)	$[N, 128, 1]$	9,504
Linear (20)	$[N, 17]$	2,193
Linear (21)	$[N, 43]$	5,547
Total		17,244
Single Model		
Layer (Type)	Output Shape	# Trainable Parameters
Feature Extraction Backbone (1–18)	$[N, 128, 1]$	9,504
Linear (20)	$[N, 731]$	94,299
Total		103,803
H. Cascade Model I (Gesture Model)		
Layer (Type)	Output Shape	# Trainable Parameters
Feature Extraction Backbone (1–18)	$[N, 128, 1]$	9,504
Linear (20)	$[N, 17]$	2,193
Total		11,697
H. Cascade Model II (Subject Model)		
Layer (Type)	Output Shape	# Trainable Parameters
Feature Extraction Backbone (1–18)	$[N, 128, 1]$	9,504
Linear (20)	$[N, 43]$	5,547
Total		15,051

0.6918 in the scenario where session 1 data were used as a test. The comparative graph of these results is demonstrated in Figure 6.

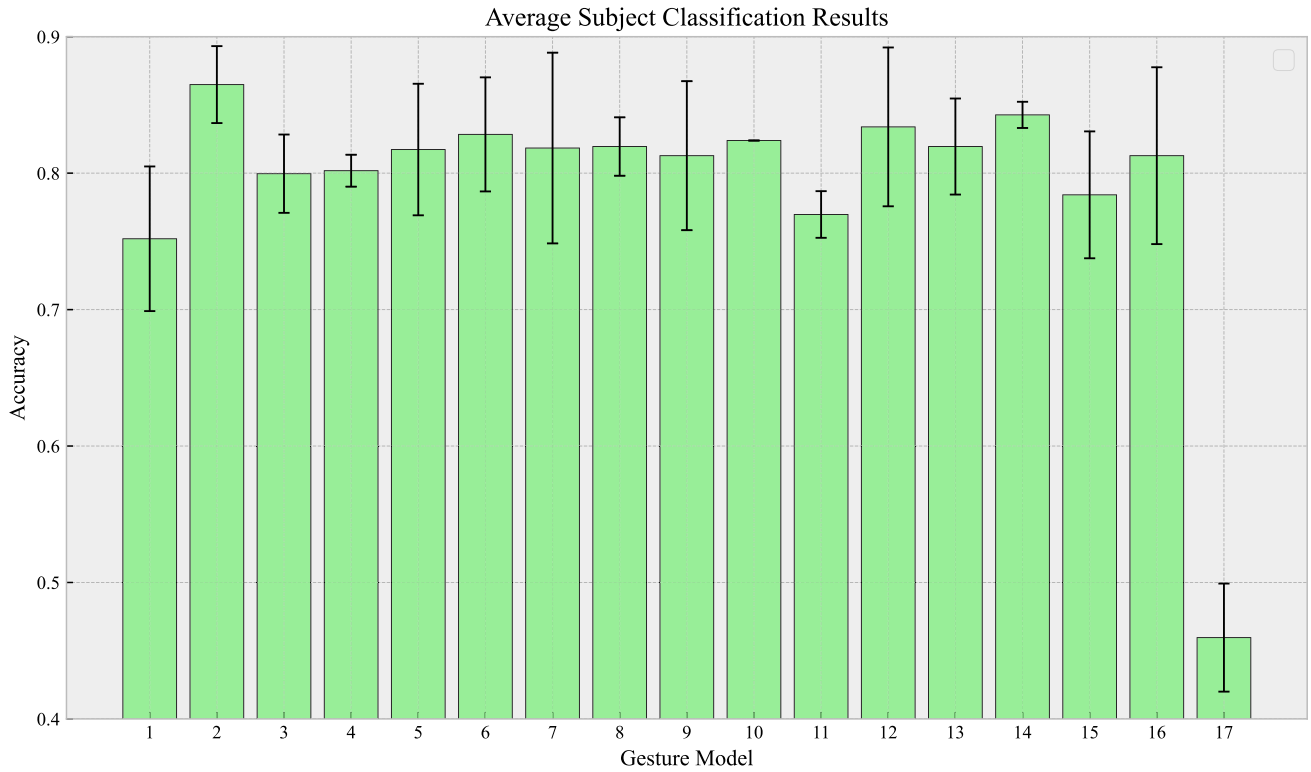


FIGURE 5. Average subject classification results for each gesture model.

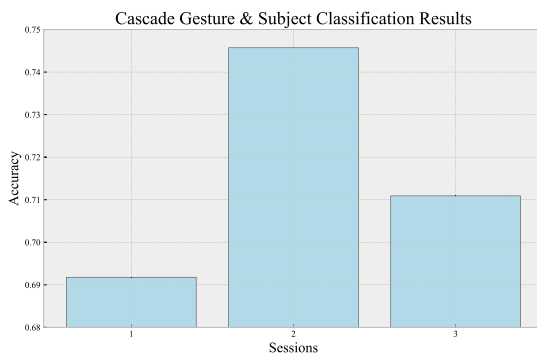


FIGURE 6. Cascade gesture and subject classification results.

4) CLASSIFICATION BASELINES RESULTS

When the classification results of different baseline models were examined from Table 5, it was observed that the hierarchical cascade model was superior to the other models with an average accuracy of $0.7162 (\pm 0.03)$ and F1-score of $0.6878 (\pm 0.03)$. It was followed by the single model structure, which achieved an accuracy of $0.6428 (\pm 0.03)$ and F1-score of $0.6428 (\pm 0.03)$. Finally, it was observed that the multitask model achieved an accuracy and F1-score performance of $0.5025 (\pm 0.05)$ and $0.4590 (\pm 0.05)$, respectively. The highest score obtained among all experiments was determined to be obtained in session 2 of

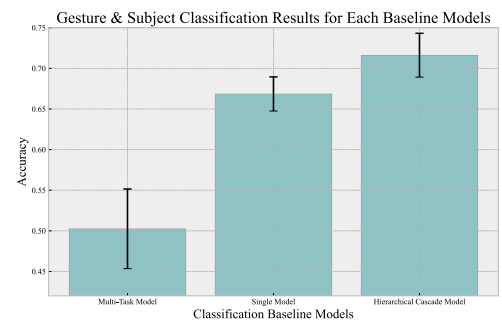


FIGURE 7. Gesture and subject classification results across various baseline models.

the hierarchical cascade model with an accuracy of $0.7457 (\pm 0.03)$ and an f1-score of $0.7213 (\pm 0.03)$. The average of these results is shown in Figure 7.

5) RESULTS OF CHANNELS FROM DIFFERENT REGIONS

In order to measure the effect of different channel regions on classification during training, classification was also performed with signals received only from the wrist and forearm regions. When the results obtained were examined (see Table 5), average accuracies of $0.3452 (\pm 0.01)$, $0.4316 (\pm 0.05)$ and $0.5025 (\pm 0.05)$ were obtained for the wrist region, the forearm region, and all channels, respectively, in multi-task learning. In the single-model structure, these

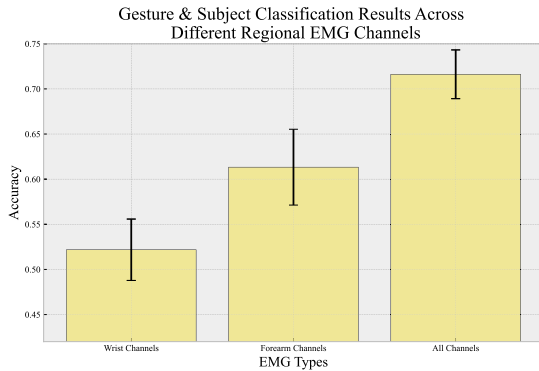


FIGURE 8. Classification performance of gestures and subjects across regional EMG channels.

accuracy values were measured as 0.5034 (± 0.03), 0.5781 (± 0.07) and 0.6685 (± 0.02), respectively. When the results of the Hierarchical Cascade model training were examined, 0.5218 (± 0.03) accuracy was obtained with signals collected from the wrist region, 0.6133 (± 0.04) with the forearm region channels, and 0.7162 (± 0.03) accuracy was obtained in the training where all channels were used. A comparison of these results is presented in Figure 8.

B. DISCUSSION

Upon examining the model results, it is observed that hand gesture classification demonstrates greater discriminative power compared to subject classification. As a result, the accuracy performance for gesture recognition is notably higher. The reasons behind this can be interpreted as the low number of classes and the patterns formed in the EMG signals having low variation differences between individuals and high variation differences between gestures. The situation where these variation differences can be interpreted most clearly is in the “rest” class. Accordingly, the “rest” class has the highest classification accuracy due to the absence of any movement compared to other gestures, and the classification performance remained quite low compared to other gesture models due to the low variation among participants in the model belonging to this gesture.

The leave-one-session-out strategy was applied while evaluating the model. In this way, in the training and test separation, two session data were separated as training and one session data was separated as test for each training. In this respect, it was observed that the most difficult evaluation session was the first general biometric session for gesture and the third session for gesture classification. It is possible to say that the best classification was in the second session. The reason for this may be that the second session is in the middle compared to the other two experiments. These experiments were applied at certain intervals, so it is possible to say that anatomical and physiological changes are possible depending on time. The experimental application, which is in the middle of these changes, can carry more information that is common to both experiments. For this reason, it is thought that this

may have an effect on the success of the second experimental session.

1) GESTURE COMPARISON IN SUBJECT CLASSIFICATION

The dataset employed during the simulation contains various human activities. Owing to this diversity, it is possible to analyze which gestures are more effective in distinguishing people. An examination of the results in Table 3 reveals that the data associated with gesture 2 exhibits a high average performance in subject classification. When Table 1 is examined, it can be observed that this hand gesture is defined as thumb abduction (TA). TA movement is one of the key findings of this study, as it demonstrates a high level of discrimination to classify the subject and thus produces strong performances. This movement can be expressed by the force applied with the thumb to a button in the palm. This can be evaluated in various behavioral studies such as EMG measurement, applied pressure and application rhythm during the performance of this movement in future biometric system studies. In addition, this movement is relevant to biometric systems that analyze how a person grasps objects. For example, users may be asked to grasp various items, such as a door handle, and access can be granted based on behavioral biometric analysis, such as the way they open a locked door. The following movement with strong classification performance is obtained with the forearm pronation (FP) gesture. FP is directly caused by forearm muscle activity, and strong results may have been obtained because the measurements were taken from this region. An examination of the “rest” condition, which yielded the least successful results, suggests that the absence of movement prevents the detection of meaningful activity during measurement.

2) CLASSIFICATION BASELINE COMPARISON

It has been observed that the presented cascade structure performs better than single and multi task learning. The lowest classification performance was obtained with multi task learning. This learning has a structure that originates from a common backbone and is divided into two branches in the last step, which creates lower costs compared to the single model in terms of the number of learnable parameters (see Table 6 and Table 7). Since, the single model tries to separate into 731 classes in the classification layer, while multi task classifies two independent (43 subjects and 17 gestures) in parallel in the last layer. Although this situation makes the multi task model advantageous, one of the disadvantages of having a common backbone is that it creates confusion in the model about whether the model focus should be on the difference between gestures or who that gesture belongs to, and this negatively affects the performance. On the other hand, the single training approach established this differentiation from the beginning by clearly identifying which gesture is associated with each individual through a single class. This strategy minimizes the number of scenarios the model needs to concentrate on,

allowing for a more intense focus on variations. However, the variation differences between gestures could not be prevented from being more dominant than the variation differences between subjects, and this situation revealed the necessity of learning gesture differences and subject differences separately. For this reason, the necessity of applying two different learning strategies with a cascade structure in a hierarchical manner was revealed. This model initially conducts gesture classification with significant variation differences, subsequently invoking the designed model architecture corresponding to the estimated gesture class to execute the classification of the subject associated with that gesture. As a result of this process, two different outputs are created (gesture and subject) and it is possible to determine which gesture was made and to whom it belonged. This structure provides the model first focusing on gesture differences and then on subject differences in the relevant gesture. It has been observed that with this baseline model structure, a performance improvement is achieved compared to other baseline models. This improvement was also found to be statistically significant according to the Mann–Whitney U test, with p-values (according to both F1 and Accuracy scores of each session) of 0.022 (compared to Multi-Task Model) and 0.0260 (compared to Single Model), both below the 0.05 threshold.

The proposed hierarchical cascade model is highly suitable for real-time deployment, achieving average inference latency 32.37 ± 4.59 ms per sample with low memory and power consumption. Due to its lightweight first stage (with ~ 11 k trainable parameters) and conditional execution strategy, the model significantly reduces computational load and storage requirements, while maintaining high accuracy even on edge devices (see Table 7 for detailed architecture design).

3) REGIONAL EMG CHANNELS COMPARISON

The EMG signal data used in the study are of two types, 28-channel forearm and wrist regions. This channel information was considered as ablation standard. Accordingly, the total channel information of 28 channels was created in two separate studies as channels belonging to wrist and forearm. The models used in the experiment were carried out with 12 channels collected from the wrist region and 16 channels collected from the forearm region. Accordingly, when the obtained results were examined, it was observed that the highest classification of the wrist was 56.09% in the Cascade model as in the previous scenario, and the highest score of the forearm was 66.19% in the similar model strategy. In the light of these results, it can be observed that the information obtained from the forearm is more important in gesture and subject classification than the information obtained from the wrist. From an anatomical point of view, the vast majority of hand and finger movements are controlled by the muscles in the forearm region [37]. Finger and hand movements such as bending and twisting are performed with the activity of flexor and extensor type muscles located in this region. The movement created here is transmitted to the

fingers with the help of the tendons located in the wrist region. Therefore, the fact that the wrist region is poor in terms of muscle tissue compared to the forearm can cause problems in obtaining sufficient information in EMG signals measuring the potential tension of the muscles. In addition, the joints located in this region can cause interferences, called artifacts, during measurement and create additional noise in the signal. This distortion was observed to cause classification errors in cases where only wrist information was used in the signal classification stage applied in the study. On the other hand, it is observed that the performance increases in the 28-channel structure where both information is used. This finding demonstrates that using both information together yields stronger and more meaningful results; statistically, the p-values (conducted using the Mann–Whitney U test according to both F1 and Accuracy scores of each session) were calculated as 0.0022 (compared to wrist region channels) and 0.0043 (compared to forearm region channels), both below 0.05, indicating significance.

Despite all the advantages of receiving data from both regions in a multi-modal manner, the use of too many electrodes in terms of design and ease of use creates a problem that should be considered in the placement of these electrodes. It is clear that the realization of a wrist-based model will be quite advantageous in these respects. For example, a wearable technology product can be created with a wristband design. For this reason, in future studies, it can be considered to increase performance with data collected from the wrist as a priority manner.

V. CONCLUSION

Electrical activity occurs during muscle movements when motor neurons stimulate muscle fibers. This electrical activity can be measured using surface EMG. Since the EMG device reflects muscle activities, it has become one of the preferred control methods in applications such as various prosthetic arm designs or brain-computer interface systems. By measuring the EMG, hand gestures can be detected through the activation of muscle fibers in certain patterns. In addition, EMG information has been used in identifying a person and especially in biometric system designs thanks to its repeatable, person-specific features found in each individual. In this study, two different classifications, hand gesture classification and biometric identification, were used and the person and the gestures made by the person were detected with a hierarchical cascade structure. This established system achieved successful performance compared to different baseline structures. In addition to the successful results obtained, a highly efficient model was developed with around 11k parameters for gesture classification, approximately 15k for subject identification, and a total of ~ 26 k parameters, achieving a very low inference latency of ~ 32 ms per sample. In future studies, a comprehensive data set can be established encompassing a broader population of various races, allowing experiments to be conducted on this dataset to facilitate a more detailed analysis. It is thought that this established

system will pave the way for EMG-based biometric system designs that can be developed in the future.

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BERKE CANSIZ received the B.Sc. degree in biomedical engineering from Yıldız Technical University, in 2024, where he is currently pursuing the M.Sc. degree with the Department of Electronics and Communication Engineering. His research interests include biomedical signal and image processing, machine learning, and biometric system design.



HATICE VILDAN DUDUKCU received the B.Sc., M.Sc., and Ph.D. degrees in electronics and communication engineering from Yıldız Technical University. She began her academic career as a Research Assistant, in 2016, and is currently an Assistant Professor with the Department of Electronics and Communication Engineering, Faculty of Electrical and Electronics, Yıldız Technical University.



MURAT TASKIRAN received the B.Sc., M.Sc., and Ph.D. degrees in electronics and communication engineering from Yıldız Technical University. He began his academic career as a Research Assistant, in 2014, and later served in various academic roles, including Lecturer. Since 2023, he has been an Assistant Professor with the Department of Electronics and Communication Engineering, Faculty of Electrical and Electronics, Yıldız Technical University.



NIHAN KAHRAMAN received the B.Sc., M.Sc., and Ph.D. degrees in electronics and communication engineering from Yıldız Technical University, Istanbul, Türkiye, in 2001, 2003, and 2008, respectively.

She was the Deputy Head of the Department, from 2017 to 2022. She is currently an Associate Professor with the Department of Electronics and Communication Engineering, Yıldız Technical University. She is actively involved in research projects and scientific activities, making significant contributions to the fields of machine intelligence and electronic design automation. Her research interests include biometric system design, analog integrated circuits, and machine learning applications in electronics. She has supervised several master's and Ph.D. theses and has published numerous scientific papers in her areas of expertise.

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